SpecialistRoute Handover — 4. Configs

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Every setting lives in section 2 of `PlantDoc_Main.ipynb`: `CONFIG_BASE` holds what all runs share, `COMPONENTS` holds one preset per component, and `apply_component(name)` resets `CONFIG` from both before each run.

Shared settings (CONFIG_BASE)

Setting	Value	Meaning
`dataset_root`	`/home/auc/PlantDoc/Datasets/dataset`	Final train / val / test
`derived_root`	`/home/auc/PlantDoc/Datasets/derived_v2`	Auto-built specialist and classifier folders
`output_dir`	`/home/auc/PlantDoc/Experiments`	All results; one folder per run family
`experiment_prefix`	`v2`	Experiment folder = `v2_<Component>`
`checkpoint_monitor`	`val_loss`	Best checkpoint chosen on the held-out validation set
`expected_test_count`	236	`check_protocol()` fails if the test set differs
`multi_seed_list`	42, 123, 2024, 779, 1027	The 5 seeds behind every reported mean
`batch_size`	32	
`reduce_lr_factor` / `_patience` / `_min_lr`	0.5 / 30 epochs / 1e-6	ReduceLROnPlateau on val_loss
`early_stopping`	`False`	Every run trains its full epoch budget
`use_augmentation`	`True`	Flips and standard online augmentation
`use_cutmix` / `cutmix_alpha`	`True` / 1.0	
`use_mixup` / `mixup_alpha`	`True` / 0.2	
`use_clahe`	`False`	Images are already CLAHE-processed on disk

Setting	Value	Meaning
`use_focal_loss`, RandAugment, GridMask, copy-paste	`False`	Tried in v1, off in v2
`save_model`	`True`	Required: the best checkpoint is reloaded for evaluation

Per-component presets (COMPONENTS)

Every component uses 384×384 inputs and ImageNet-pretrained weights.

Component	Model	Epochs	LR	Fine-tune flag	Unfreeze at epoch	Fine-tune LR	Class folders matched
Classifier	ConvNeXt-Large	140	7e-5	True	50	5e-5	All, grouped by species (7 classes incl. Rest)
Apple	ConvNeXt-Large	100	1e-4	False	—	—	"Apple"
Bell pepper	ConvNeXt-Base	140	1e-4	True	40	5e-5	"Bell_pepper"
Corn	ConvNeXt-Base	100	1e-4	True	40	5e-5	"Corn"
Potato	ConvNeXt-Base	100	1e-4	True	40	5e-5	"Potato"
Tomato	ConvNeXt-Large	300	1e-4	False	—	—	"Tomato"
Grape	ConvNeXt-Base	50	1e-4	True	20	5e-5	"Grape"
Rest	ConvNeXt-Large	200	5e-5	False	—	—	Blueberry, Cherry, Peach, Raspberry, Soyabean, Squash, Strawberry

Read before relying on the fine-tune columns: the fine-tuning callback very likely never unfreezes the backbone under Keras 3, so the "True" rows may in practice be frozen-backbone models whose learning rate drops at the unfreeze epoch. Verify this first (Known Issues).

Potato changed in v2: it was EfficientNet-B0 Noisy Student at 484 px. Those weights did not load under Keras 3 and training stalled at loss ≈ ln 2, so Potato now uses the Corn preset (ConvNeXt-Base). Disclose this in the paper.

Pipeline and ensemble evaluation settings

These settings have no single config key. They were fixed on validation, then used once on test.

Setting	Value	How it was decided
Router	7-class classifier (6 crops + Rest), no confidence threshold	v1: +18.5 pp over a 6-class router with a threshold (182 vs 142 / 236, seed 42)
Routing	Hard: each image goes only to its top-predicted species	Soft routing gained 1 validation image; the rule fixed in advance keeps the simpler option
Ensemble	Average the softmax of all 5 seeds, router and specialists	Beats every single seed on validation and on test
Test-time augmentation	Off	Horizontal-flip TTA changed nothing on validation
Image loading	TensorFlow decode + bilinear resize	Identical to training; PIL changed results in v1
Checkpoints used	`v2_<Component>/<model>_seed<seed>.keras`	Best validation-loss epoch of each run

Tie rule, fixed before any test result: when two options differ by 1 validation image or less, keep the simpler one.

Changing settings safely

- **Never edit `CONFIG` directly between runs.** Change `CONFIG_BASE` or `COMPONENTS`, or pass overrides: `apply_component("Tomato", epochs=10)` for a quick smoke test. `apply_component` resets `CONFIG` first, so nothing leaks from the previous run in the same kernel.
- **Overrides must be existing keys.** A misspelt key raises `KeyError` instead of being silently ignored.

- **Changing a preset invalidates old results.** The orchestrator (section 9a) compares each component's saved `config.txt` with its current preset (model, image size, epochs, learning rates, fine-tuning, batch size, LR schedule, CutMix/MixUp). On any difference it moves the old results, and all pipeline evaluations, to `Experiments/v2_superseded/` and retrains that component for every seed.
- **Checkpoint paths are never overwritten.** `get_model_save_path` raises `FileExistsError`; the multi-seed runner moves a half-finished checkpoint aside before retraining.

Where the presets came from

The presets were carried over from the final v1 configurations (`Plantdoc.csv`, classifier exp_66, Rest exp_122). In v1, configurations were compared on the test set, so the v2 presets still carry a small amount of that optimism. Disclose it as a limitation. Re-tuning on validation would remove it, but means retraining everything.